

JEE MAIN REVISION MASTER: COMMUNICATION SYSTEMS

Subject: Physics | Target Scope: JEE Main Previous Years | Empirical Base: 2 Analyzed PYQs

1. High-Yield Concept Priority Matrix

Priority Tier	Concept Name	Frequency	Category
Standard	Modulation & Space Wave Propagation Range	2 PYQs	Communication

2. Master Formula Reference Card

Concept / Topic	Mathematical Formulation & Governing Equations
Modulation & Space Wave Propagation Range	$\mu = \frac{A_{\max} - A_{\min}}{A_{\max} + A_{\min}}, \quad d = \sqrt{2Rh}, \quad \text{Population Covered} = \rho \pi d^2 = \rho \pi (2Rh)$

3. Deep-Dive Concepts & Linked PYQ Evidence

3.1 Modulation & Space Wave Propagation Range (Standard Yield — 2 PYQs)

Core Principle: Modulation index in AM: $\mu = (A_m / A_c) = \frac{A_{\max} - A_{\min}}{A_{\max} + A_{\min}}$ ($0 \leq \mu \leq 1$ to prevent distortion). Bandwidth of AM wave is $2f_m$. Maximum line-of-sight range for a transmitting antenna of height h_t : $d = \sqrt{2Rh_t}$. Between transmitter and receiver: $d_{\max} = \sqrt{2Rh_t} + \sqrt{2Rh_r}$.

■ **Caution / JEE Pitfall:** Antenna range d depends on $\sqrt{h}$ (square root of height). To double the transmission range, the tower height must be quadrupled (4×)!

■ **Speed Trick:** Using Earth radius $R = 6.4 \times 10^6$ m: $d \approx \sqrt{2 \times 6400000 \times h} = \sqrt{1.28 \times 10^7 h} \approx 3578\sqrt{h}$ meters.

Representative Source PYQs:

- [2018 Shift] Q#1 (MCQ): "The radius of the Earth is 6400 km . If the height of an antenna is 500 m , then its range is..." [Ans: N/A]
- [2017 Shift] Q#2 (MCQ): "A basic communication system consists of (i) Transmitter (ii) Information sources (iii) User of information (iv) Channel (v) Receiver..." [Ans: N/A]

4. Benchmark Illustrative PYQs with Master Solutions

Benchmark Problem 1: KCET 2018 (Q#1)

The radius of the Earth is 6400 km . If the height of an antenna is 500 m , then its range is

- (A) 800 km
- (B) 100 km
- (C) 80 km
- (D) 10 km

Correct Answer: None

Solution Analysis: To find the range of the antenna (i.e., the distance to the horizon), we can use the formula: $d = \sqrt{2Rh}$ where: R is the radius of the Earth h is the height of the antenna Since the Earth's radius is given as 6400 km and the antenna's height is 500 m, we first convert the height into kilometers: $500\text{m} = 0.5\text{km}$ Now, plug the values into the formula: $d = \sqrt{2 \times 6400 \times 0.5} = \sqrt{6400} = 80\text{km}$ Thus, the range of the antenna is 80 km, which corresponds to Option C.